Computer Science - Graduate Level Assessment 12 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following is true of interrupts?
   (a) They are generated when memory cycles are "stolen".
   (b) They are used in place of data channels.
   (c) They can indicate completion of an I/O operation.
   (d) They cannot be generated by arithmetic operations.

2. How is IP address spoofing detected?
   (a) Installing and configuring a IDS that can read the IP header
   (b) Comparing the TTL values of the actual and spoofed addresses
   (c) Implementing a firewall to the network
   (d) Identify all TCP sessions that are initiated but does not complete successfully

3. You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?
   (a) No, I cannot compute the key.
   (b) Yes, the key is k = m xor c.
   (c) I can only compute half the bits of the key.
   (d) Yes, the key is k = m xor m.

4. A hash function guarantees the integrity of a message. It guarantees that the message has not be
   (a) Replaced
   (b) Overview
   (c) Changed
   (d) Violated

5. What are the port states determined by Nmap?
   (a) Active, inactive, standby
   (b) Open, half-open, closed
   (c) Open, filtered, unfiltered
   (d) Active, closed, unused

True / False (10 marks)

Answer True or False

6. True or False: John the Ripper is primarily designed for network intrusion detection and real-time traffic analysis.
7. True or False: CFB is a standard block cipher operating mode commonly used in encryption algorithms.
8. True or False: Snort is primarily designed for network analysis in multiprotocol environments rather than intrusion detection.
9. True or False: ARPANET is primarily known for enabling anonymous file transfers on the darknet.
10. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is approximately 1/K.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to search a literals string in a string and also find the location within the original string where the pattern occurs by using regex.
12. Write a function to find t-nth term of geometric series.

End of Paper